

Simulation Reliance Boundary Module (SRBM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS)

Module: SRBM — Simulation Reliance Boundary Module

Operational Layer: Simulation Reliance Boundary Layer

Category: Governance & Enforcement

Subcategory: Simulation Governance

Type: Core Module

Governed Space: Simulation Reliance Boundary

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS® · INTEGROS® · ADIT® · AGA™ ·
AIG® · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodology Intellectual Property Protection

Canonical Language: English

Canonical Definition

Simulation Reliance Boundary Module (SRBM) is the third Core Module of the Simulation Evidence & Reliance Boundary Standard (SERBS) within SIMULOS® — Simulation Governance Architecture.

SRBM governs the maximum legitimate simulation-specific reliance that may be placed upon qualified and weighted Simulation Evidence by defining the objects, purposes, configurations, conditions, scenarios, parameter ranges, environments, temporal contexts, propositions, decisions, and forms of use within which reliance remains supported by the governed simulation basis.

The Module establishes the methodological boundary between:

what simulation evidence legitimately supports

and

what would constitute reliance beyond that support.

SRBM does not determine whether the total evidence basis is sufficient for a broader Validation Determination, nor does it authorize deployment, operational use, or organizational decisions.

Its task is narrower:

to determine how far reliance upon simulation-derived evidence may legitimately extend before that evidence is being used beyond the governed basis that gives it meaning.

Core Question

How far may reliance upon simulation-derived evidence legitimately extend before the evidence is being used beyond its governed basis?

Architectural Position

SRBM is the third module within SERBS.

It follows:

SEQM — Simulation Evidence Qualification Module

→ determines what legitimately qualifies as Simulation Evidence

and:

SEWM — Simulation Evidence Weight Module

→ determines how much simulation-specific evidential significance that evidence may carry.

SRBM then establishes:

where legitimate reliance upon that evidence stops.

It precedes:

SUM — Simulation Use Limitation Module

which governs explicit restrictions, exclusions, caveats, unsupported uses, and prohibited interpretations associated with that reliance boundary.

The sequence is therefore:

Qualification → Weight → Reliance Boundary → Use Limitation.

Internal Flow Position

Qualified Simulation Evidence Simulation Evidence Weight → Intended Reliance Identification → Reliance Purpose Definition → Evidence-to-Reliance Alignment → Reliance Condition Identification → Boundary Constraint Assessment → Reliance Expansion Assessment → Unsupported Reliance Detection → Reliance Boundary Determination → Simulation Reliance Boundary State → Governed Simulation Reliance Boundary → SUM — Simulation Use Limitation Module

Operational Space

SRBM governs the space between:

evidential significance

and

permitted simulation-specific reliance.

This operational space includes: intended reliance; reliance purpose; reliance object; reliance proposition; reliance context; reliance conditions; object-specific reliance; configuration-specific reliance; scenario-specific reliance; parameter-specific reliance; environment-specific reliance; temporal reliance; conditional reliance; comparative reliance; predictive reliance; extrapolative reliance; reliance expansion; unsupported reliance; reliance boundary crossing; reliance degradation; reliance suspension; reliance reassessment.

Operational Scope

SRBM applies whenever simulation-derived evidence is expected to influence or support: engineering judgment; design decisions; scientific conclusions; safety arguments; mission planning; mission decisions; system qualification; comparative evaluation; performance assessment; scenario selection; infrastructure planning; operational preparation; risk analysis; AI evaluation; autonomous-system evaluation; deployment preparation; validation processes; regulatory evidence packages; investment or resource decisions; other material forms of reliance.

The Module applies regardless of whether the evidence originates from: one simulation; repeated simulations; Monte Carlo studies; scenario portfolios; large simulation campaigns; digital twins; engineering simulations; mission simulations; safety simulations; scientific simulations; physical simulations; hybrid simulations; AI simulations; autonomous-system simulations; future simulation technologies.

SRBM remains technology-neutral.

Its concern is not:

How advanced is the simulation?

Its concern is:

How far may the evidence produced through that simulation legitimately be relied upon? Evidence Weight $\neq$ Reliance Boundary

SRBM enforces the distinction:

High Evidence Weight $\neq$ Broad Reliance

Evidence may carry substantial evidential weight for a narrowly defined proposition while providing little or no legitimate basis for broader generalization.

For example:

A simulation campaign may provide substantial evidence that:

Configuration A remains stable under Scenario Family S within Parameter Range P.

That does not automatically support:

Configuration A will remain stable under all operational conditions.

SEWM determines evidential strength.

SRBM determines the extent of legitimate reliance.

Reliance Must Be Explicit

Reliance must not remain implicit.

SRBM requires the intended form of reliance to be identified before its legitimacy can be assessed.

The intended reliance should establish: what decision, proposition, comparison, judgment, or action is being informed; which Simulation Evidence objects are being relied upon; what evidence weight those objects carry; which Simulation Object is concerned; which object version and configuration apply; which represented conditions apply; which scenario space applies; which uncertainty remains; what temporal context applies; what extension beyond directly simulated conditions is proposed.

Without an explicit intended reliance, a reliance boundary cannot be meaningfully governed.

Reliance Object Principle

Simulation evidence may be relied upon only in relation to a sufficiently defined Reliance Object.

The Reliance Object may be: a system; a component; a configuration; a design; a process; a mission phase; an operational concept; a simulated behavior; a comparative alternative; a parameter range; another explicitly bounded subject.

The Reliance Object must remain aligned with the Simulation Object and evidential basis.

Evidence concerning one object must not silently migrate to another.

Reliance Purpose Principle

The legitimacy of reliance depends upon why the evidence is being used.

The same evidence may legitimately support:

exploratory engineering

while not supporting:

safety-critical deployment.

It may support:

comparison between two simulated configurations

while not supporting:

absolute prediction of real-world performance.

It may support:

mission contingency analysis

while not independently supporting:

mission readiness determination.

SRBM therefore requires reliance to remain purpose-specific.

Reliance Context

Every governed reliance boundary should identify the context within which reliance remains supported.

Context may include: Simulation Object; object version; object configuration; simulation purpose; simulation scope; represented reality; abstraction level; fidelity; assumptions; model version; environment configuration; input basis; scenario family; parameter range; execution basis; output class; inference basis; correlation basis; uncertainty; temporal applicability; evidence weight.

The boundary is therefore multidimensional.

It is not merely a statement that evidence is:

usable

or

not usable.

Object Boundary

Simulation evidence must remain bounded to the object for which its evidential meaning was established.

Evidence derived from:

Object Version V3

must not automatically be relied upon for:

Object Version V4

where the change may materially affect the evidence.

Similarly:

Component Evidence ≠ System Evidence

unless a governed basis supports that extension.

Configuration Boundary

Evidence may apply only to a specific configuration.

Configuration differences may include: hardware; software; model parameters; architecture; sensor configuration; actuator configuration; material properties; mission configuration; operating mode; system integration state.

A materially changed configuration may require reliance reassessment.

Scenario Boundary

Evidence produced within a defined scenario space must not automatically be generalized outside that scenario space.

SRBM requires distinction between:

Tested Scenario Space

and

Untested Scenario Space.

Scenario coverage may be broad without being universal.

Therefore:

Scenario Coverage ≠ Universal Coverage.

Parameter Boundary

Simulation evidence may be valid only within defined parameter ranges.

Examples include: velocity; temperature; pressure; load; mass; latency; sensor range; environmental condition; resource availability; probability distribution; behavioral parameter.

Reliance beyond tested or governed parameter ranges constitutes extrapolation and requires explicit justification.

Environment Boundary

Evidence may be dependent upon the environment represented by the simulation.

Environmental boundaries may include: terrain; weather; atmospheric conditions; lighting; traffic; infrastructure; physical interaction; human behavior; electromagnetic conditions; gravity; thermal conditions; communication availability.

Evidence derived from one represented environment must not automatically be generalized to materially different environments.

Temporal Boundary

Simulation evidence may have a legitimate reliance period.

Reliance may degrade where: the Simulation Object changes; operational reality changes; input distributions change; assumptions become obsolete; models are updated; new failures emerge; new evidence contradicts earlier results; simulation-to-reality correlation changes; relevant scenarios evolve.

SRBM therefore recognizes:

Evidence can remain historically correct while becoming operationally stale.

Correlation Boundary

Where reliance concerns real-world behavior, SRBM must consider the governed simulation-to-reality correlation basis established through SRCS.

Strong correlation under:

Conditions A, B, and C

does not automatically establish equivalent correlation under:

Conditions D, E, and F.

Therefore:

Correlation within a tested domain $\neq$ correlation outside that domain.

Uncertainty Boundary

Reliance must reflect material uncertainty.

Uncertainty may constrain: the confidence of a conclusion; the parameter range; the scenario range; the temporal applicability; the permitted decision class; the degree of extrapolation; the strength of predictive claims.

SRBM does not require uncertainty to eliminate reliance.

It requires uncertainty to constrain reliance where materially relevant.

Direct Reliance and Extended Reliance

SRBM distinguishes between: Direct Reliance

Reliance that remains within the object, conditions, scenarios, parameters, purposes, and propositions directly supported by the governed evidence.

Extended Reliance

Reliance that moves beyond one or more dimensions directly supported by the evidence.

Extended reliance is not automatically prohibited.

But it requires an explicit governed justification.

Reliance Expansion

Reliance Expansion occurs when evidence is used beyond its established boundary.

Expansion may involve: another object; another configuration; another scenario; another environment; another parameter range; another time period; another decision purpose; another operational condition; another population; another mission phase; a broader proposition.

SRBM requires expansion to be visible.

Reliance must not expand silently.

Reliance Expansion Test

Before reliance is expanded, SRBM should determine:

1. What boundary dimension is being crossed? 2. How materially different is the new context? 3. Which simulation assumptions remain applicable? 4. Which representation conditions remain applicable? 5. Does the model remain applicable? 6. Do the inputs remain representative? 7. Does scenario coverage support the extension? 8. Does the correlation basis transfer? 9. What new uncertainty is introduced? 10. Is additional simulation, evidence, correlation, or validation required?

Extrapolation Principle

Extrapolation is a particularly important form of reliance expansion.

Simulation evidence may be extrapolated beyond directly tested conditions only where the extrapolation basis is explicitly governed.

The extrapolation should identify: originating evidence; originating range; target range; methodological basis; assumptions; expected uncertainty; model applicability; correlation relevance; known limitations.

Extrapolation must be treated as extrapolation, not silently presented as direct evidence.

Comparative Reliance

Simulation evidence may provide strong support for comparative propositions even where it cannot support absolute real-world claims.

For example:

Configuration A performs better than Configuration B under governed Scenario Set S

may be strongly supported.

That does not necessarily establish:

Configuration A will achieve Performance Level X in operational reality.

SRBM therefore distinguishes:

Comparative Reliance

from

Absolute Reliance.

Predictive Reliance

Where simulation evidence is used predictively, SRBM requires the predictive reliance boundary to remain explicit.

Predictive reliance should consider: future condition uncertainty; representation limits; assumption stability; model applicability; input distribution shift; scenario completeness; correlation basis; temporal horizon.

A simulation may legitimately inform a prediction without guaranteeing the predicted outcome.

Reliance Boundary States

SRBM may assign a governed reliance-boundary state.

A minimum state model may include: 1. Undefined Reliance

Evidence exists, but no explicit legitimate reliance boundary has been established. 2. Narrow Reliance

Evidence supports reliance only within tightly defined conditions. 3. Conditional Reliance

Reliance is legitimate only while explicitly defined conditions remain satisfied. 4. Bounded Reliance

Evidence supports a clearly established and documented reliance domain. 5. Extended Reliance — Justified

Reliance extends beyond directly simulated conditions, but the extension has an explicit governed basis. 6. Reliance Suspended

Previously supported reliance requires reassessment because of material change, contradiction, uncertainty, or loss of applicability. 7. Unsupported Reliance

The intended use exceeds the legitimate simulation-specific evidence basis.

These states are governance classifications rather than universal measures of

scientific certainty.

Unsupported Reliance

Unsupported Reliance exists where simulation evidence is used beyond the boundary its governed basis can legitimately support.

Examples include reliance: outside simulation scope; for an excluded condition; on a materially changed object; on a materially changed configuration; outside tested parameter ranges; outside represented environments; beyond scenario coverage; beyond established correlation; after evidence becomes stale; contrary to material uncertainty; for a materially different purpose; beyond the proposition actually supported; as proof of operational performance where only simulated performance was established.

Unsupported reliance must remain identifiable even where the organization has historically relied upon the evidence in that manner.

Repeated unsupported reliance does not create legitimate reliance.

Boundary Crossing Detection

SRBM should enable organizations to detect when downstream use crosses a governed reliance boundary.

Boundary crossing may be triggered by: object change; configuration change; scenario change; environmental change; parameter expansion; purpose change; temporal change; decision-class change; new evidence; new uncertainty; correlation deterioration; contradiction; material reality change.

A detected boundary crossing should trigger:

Reliance Review

and where necessary:

Use Limitation → Additional Simulation → Additional Correlation → Validation Review → Resimulation.

Reliance Degradation

A reliance boundary may degrade over time.

Degradation may occur when: the evidence weight decreases; reality changes; the model becomes obsolete; assumptions become less defensible; scenario relevance declines; new operational observations appear; correlation weakens; contradictory evidence emerges.

SRBM therefore requires reliance boundaries to remain reviewable. Reliance Boundary ≠ Decision Authority

SRBM determines whether simulation-derived evidence supports a particular reliance domain.

It does not determine: who may make the decision; who accepts the risk; who authorizes deployment; who owns accountability; whether the organization

should act.

Those questions belong to the relevant authority, accountability, operational, risk, or validation governance structures.

Therefore:

Supported Reliance $\neq$ Authorized Decision.

SRBM Boundary with SEWM

SEWM determines:

how much evidential weight qualified Simulation Evidence carries.

SRBM determines:

how far reliance upon that evidence may extend.

Evidence weight is an input into reliance governance.

It is not the reliance boundary itself.

SRBM Boundary with SUM

SRBM establishes the legitimate reliance domain.

SUM translates that boundary into explicit: restrictions; exclusions; caveats; prohibited uses; unsupported interpretations; required conditions.

Therefore:

SRBM defines where reliance stops.

SUM defines what must not happen beyond or within that boundary.

SRBM Boundary with VALIDOS®

SRBM does not determine overall validation sufficiency or authorize operational reliance based on the total evidence basis.

Its question is:

How far may this simulation-derived evidence legitimately be relied upon as simulation evidence?

VALIDOS® asks:

Does the total validation basis sufficiently support the Validation Determination and intended reliance?

Therefore:

Simulation Reliance Boundary $\neq$ Validation Reliance Determination.

SRBM Boundary with SRCS

SRCS governs the simulation-to-reality correlation relationship.

SRBM uses that governed relationship when determining how far reality-directed reliance may extend.

It does not establish or recalculate correlation.

SRBM Boundary with ORA™

ORA™ governs actual Operational Reality.

SRBM may determine whether simulation-derived evidence legitimately supports propositions concerning a bounded part of that reality.

It does not determine operational reality itself.

SRBM Boundary with AGA™

AGA™ governs authority, responsibility, accountability, and attribution.

SRBM governs the methodological boundary of reliance upon simulation-derived evidence.

It does not assign accountability for reliance decisions. SRBM Boundary with AIG® / ASGA™

Where simulation evidence concerns AI or autonomous systems, SRBM may govern how far that simulation evidence can legitimately support propositions about the simulated system.

It does not govern the actual behavior, authority, autonomy, permissions, constraints, or operation of the deployed system.

SRBM Boundary with INTEGROS®

SRBM does not determine general integrity.

Evidence or records may possess high integrity while still having a narrow reliance boundary.

Therefore:

Integrity ≠ Breadth of Reliance.

SRBM Boundary with ADIT®

ADIT® may audit whether the evidence basis and reliance records are observable, preserved, traceable, or verifiable.

SRBM establishes the simulation-specific methodological boundary itself.

Auditability ≠ Legitimate Reliance.

Minimum Implementation Framework

A minimum SRBM implementation should establish: 1. Evidence Input

Receive qualified and weighted Simulation Evidence from SEQM and SEWM. 2. Intended Reliance

Identify the precise proposition, decision, judgment, comparison, or use for which reliance is proposed. 3. Reliance Object

Identify the object, version, configuration, or subject to which reliance applies. 4. Reliance Context

Define the relevant scenario, environment, parameter, purpose, temporal, correlation, and uncertainty conditions. 5. Evidence-to-Reliance Alignment

Determine whether the evidence directly supports the intended reliance. 6. Boundary Constraint Identification

Identify all material conditions constraining legitimate reliance. 7. Reliance Expansion Assessment

Identify whether the proposed use extends beyond directly supported conditions. 8. Unsupported Reliance Detection

Determine whether any proposed use exceeds the legitimate evidence basis. 9. Reliance Boundary State

Assign an explicit governed reliance-boundary state. 10. Reliance Boundary Record

Document the supported domain, boundary conditions, expansion basis, uncertainty, and conditions requiring reassessment.

Governance Outputs

SRBM may produce: Simulation Intended Reliance Statement Simulation Reliance Object Record Simulation Reliance Purpose Record Simulation Reliance Context Record Evidence-to-Reliance Alignment Assessment Simulation Reliance Constraint Record Simulation Reliance Expansion Assessment Simulation Extrapolation Record Simulation Reliance Boundary State Unsupported Reliance Record Reliance Boundary Crossing Record Simulation Reliance Degradation Record Simulation Reliance Boundary Statement Governed Simulation Reliance Boundary

These outputs provide the governed basis for:

SUM — Simulation Use Limitation Module.

Enterprise Application

SRBM enables organizations to answer one of the most consequential questions in simulation governance:

Where exactly does legitimate reliance on the simulation stop?

An organization may possess: millions of successful simulation runs; high-fidelity models; substantial evidence weight; strong correlation within

tested conditions; extensive scenario coverage.

Yet none of those facts independently establish unlimited reliance.

For a spacecraft, evidence concerning a governed landing envelope may not support an untested environmental regime.

For an autonomous vehicle, millions of simulated kilometers may strongly support performance within represented traffic, sensor, weather, and behavioral conditions while leaving materially different conditions outside the legitimate reliance boundary.

For a robot, evidence generated for one hardware configuration may not transfer automatically after sensors, actuators, software, payload, or operating environment change.

For an engineering system, evidence within a tested load or temperature range may not justify performance claims outside that range.

SRBM makes those boundaries explicit before narrow simulation evidence becomes broad organizational belief.

It allows organizations to distinguish:

what was simulated

from

what was evidenced

from

what may legitimately be relied upon.

Foundational Principles

Evidence weight is not breadth of reliance.

High evidence weight does not create unlimited reliance.

Reliance must remain purpose-specific.

Reliance must remain object-specific where object differences are material.

Scenario coverage does not create universal coverage.

Correlation within one domain does not establish correlation outside that domain.

Extrapolation must remain identifiable as extrapolation.

Comparative evidence does not automatically establish absolute performance.

Evidence can remain historically correct while becoming operationally stale.

Reliance expansion requires a governed basis.

Repeated unsupported reliance does not create legitimate reliance.

Supported simulation reliance does not itself constitute deployment authorization or Validation Determination.

Simulation-derived evidence must not be relied upon beyond the object, purpose, configuration, represented conditions, assumptions, parameters, scenarios, environments, correlations, uncertainties, and temporal context that establish its legitimate meaning.

Canonical Closing Statement

Simulation Reliance Boundary Module establishes the reliance-control layer within the Simulation Evidence & Reliance Boundary Standard by determining how far qualified and weighted Simulation Evidence may legitimately support a proposition, judgment, comparison, decision context, or other defined form of simulation-specific reliance. Through governance of reliance objects, purposes, contexts, configurations, scenarios, parameters, environments, temporal applicability, correlation domains, uncertainty, direct and extended reliance, extrapolation, reliance expansion, unsupported reliance, boundary crossing, degradation, suspension, and reliance-boundary states, SRBM prevents simulation evidence from silently acquiring broader applicability than the governed conditions from which its evidential meaning originates. By separating evidential weight from breadth of reliance, simulation-specific reliance from validation sufficiency, and supported reliance from decision authority, SRBM preserves the methodological boundaries of SIMULOS®, VALIDOS®, ORA™, INTEGROS®, ADIT®, AGA™, AIG®, and ASGA™ while creating the governed reliance domain required for explicit downstream use limitations. Simulation evidence may legitimately inform consequential decisions, but it must never travel beyond its governed evidential basis without an explicit and justified expansion of reliance.

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